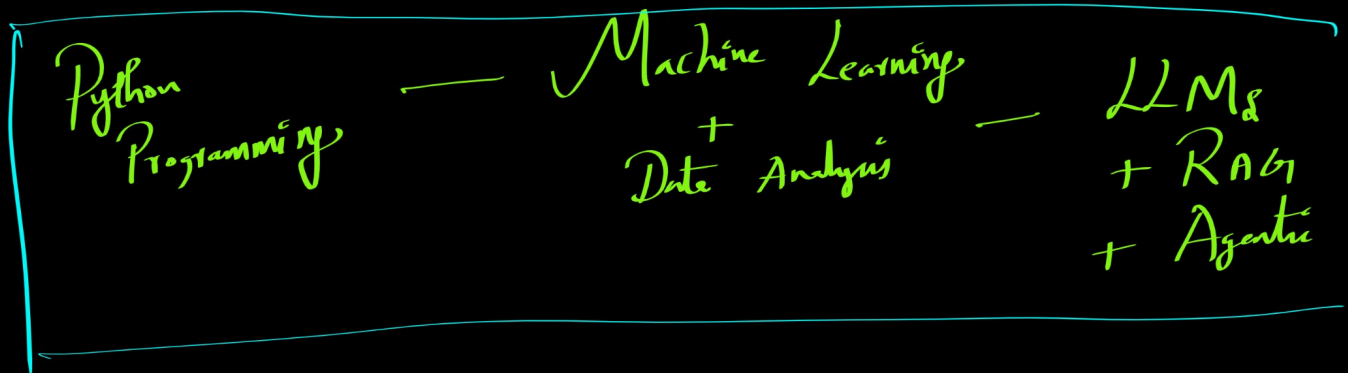
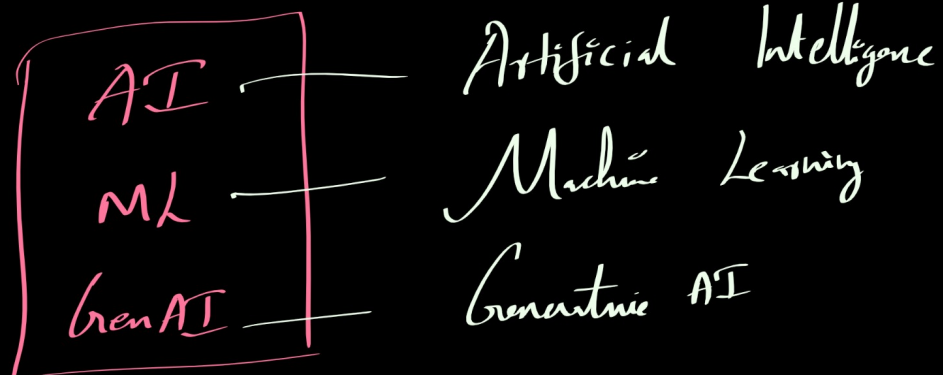


The AI Landscape, Computing & Lab Setup

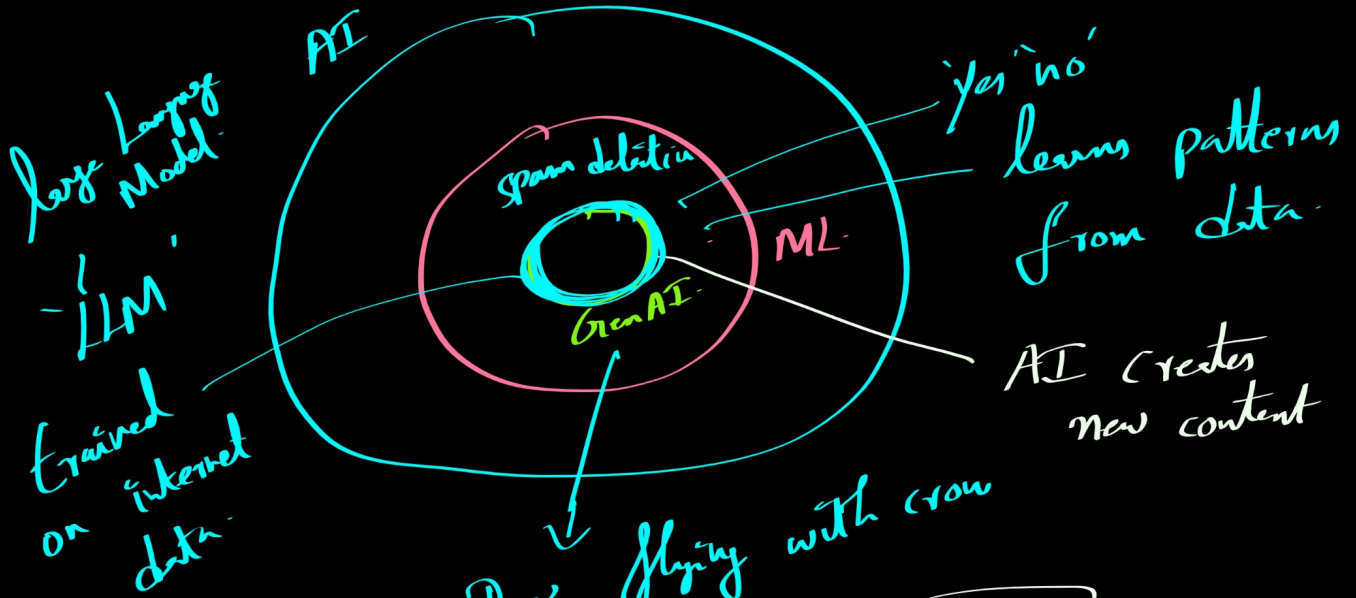


Topics to be covered:

1. AI vs ML vs Gen AI
2. History of AI ✓
3. Computer Basics ✓
Hardware vs Software
Program & Programming
Python - Programming Language
4. IDE / CLI
5. How the code runs ✓
6. VS code Installation.



AI - machine / system that mimics human brain



Dog flying with crow

Humans

Brain

Previous Data

we do something

from past Data

Brain

delivering o/p.s.

Machines

learn the pattern from the given data

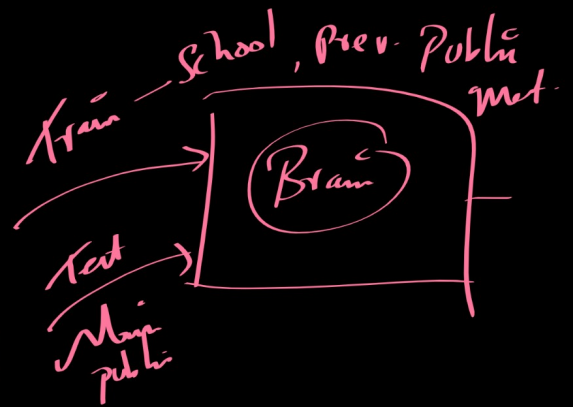
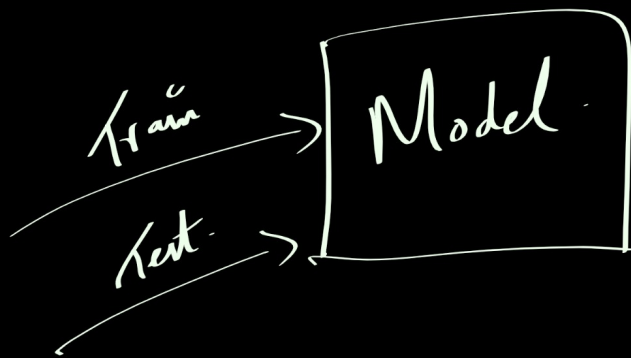
feed data

Model's

Year	Month	Season	2025
1999	May	Summer	
	June	Summer	
	Nov	Rainy	
	Dec	Winter	
2005	May	Rainy	Model



Unseen Data



Netflix — ML algorithm

Recommendations

previous data

Thousand 2

Korean Series

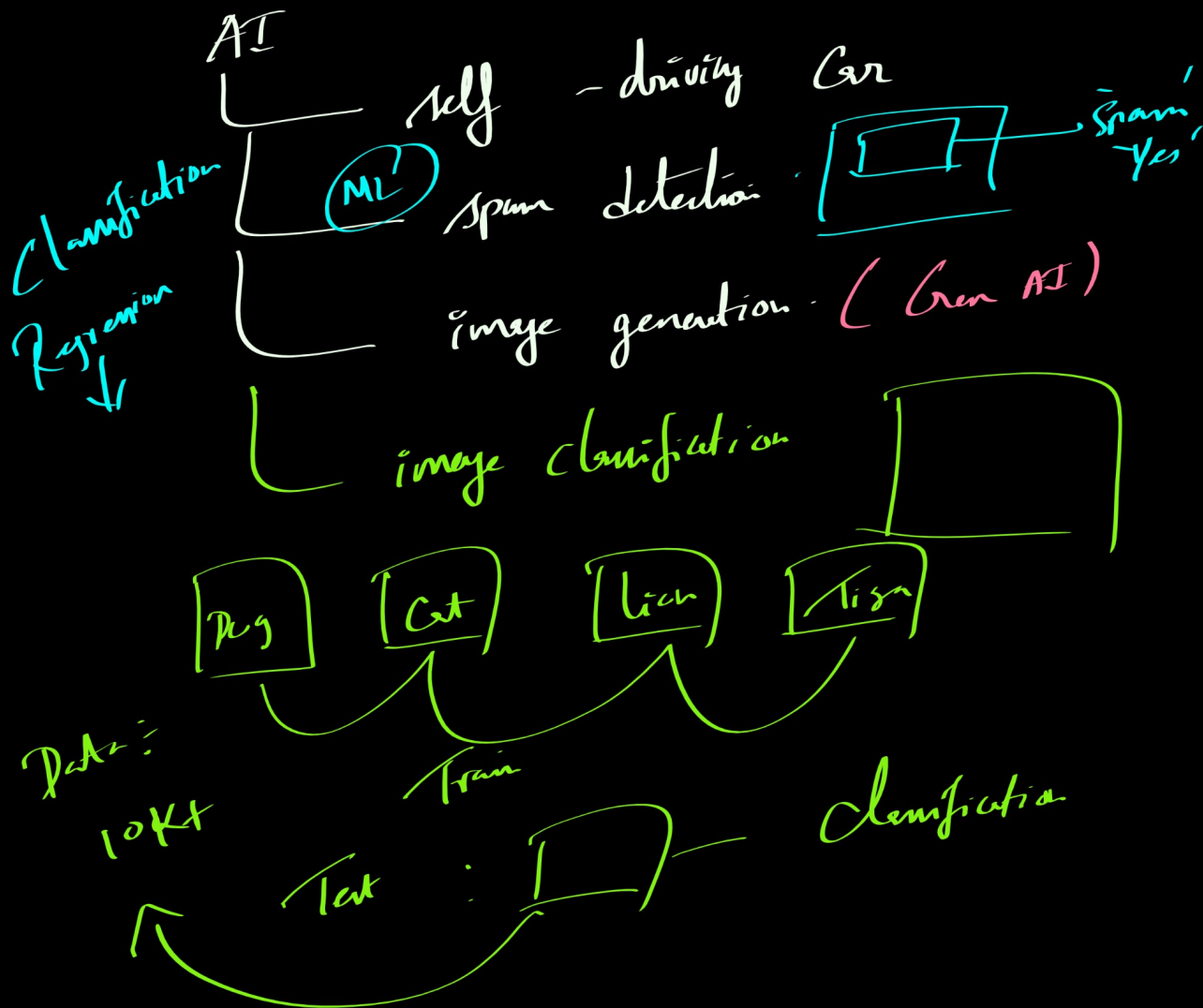
Romantic

Python Programming

↳ Cinematic Style ✓

$AI \supseteq ML \supseteq Gen\ AI$

AI - machine



RR vs ML

Correlation (0-100)

Kanhiyapuram

— 'XVK' 'DMK' 'AMK'

Classification

predicted
range of
values - Regression

'30-9' , '24-8' , '20-9'

History of AI:

Alan Turing

1950s

Birth of AI

Turing Test — A machine is intelligent if humans cannot distinguish it from a human conversation.

Can Machines think?

1956: AI — Artificial Intelligence — Dartmouth Conference.

'1960 - 1980' : Rule Based System

'Eliza' — '1966'

'if' else condition

'1960 - 1980'

if customer is sad:
'respond this'

if customer is not
satisfied:
respond this

1980+ — 'AI Winter' — Research Funding

1990s — 2000s — Internet Boom

Large collection of Data

Google	1998
Amazon	1994
Facebook	2004

Computers

2010s

How our Brain works

Neural Networks

large computational power

GPUS

Deep Learning

large models

Data

2020s: ChatGPT, Gemini, Claude

2017

Transformer Architecture

Gradient Descent

Foundation for GPT — Google

Chemistry	80	$P(\text{Chem}) = 84/100 = 0.84$ $P(\text{suber}) = 0.02$ 0.08 0.1
Chemistry	2	
Suber	8	
Mo	10	

Then

taken
 ing
 used to grade members

Playing
 play
 player

Sleeping
 sleepy
 sl

Play + ing
 sleep + ing - s
 sing + ing - s
 stinging

Computer Basics:

Hardware

Keyboard
 Mouse
 RAM
 storage device
 CPU
 Monitor

Software
 windows
 apps
 programs

Human didn't
 legs
 eyes
 body
 not there

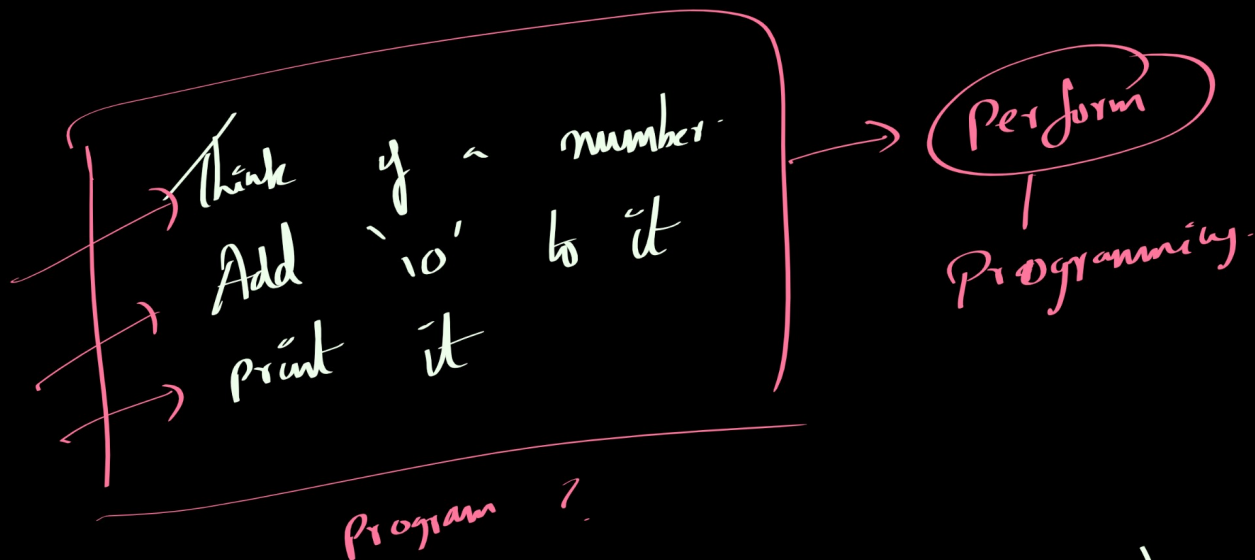
software
 Brain X
 Brain
 brain dead

Program

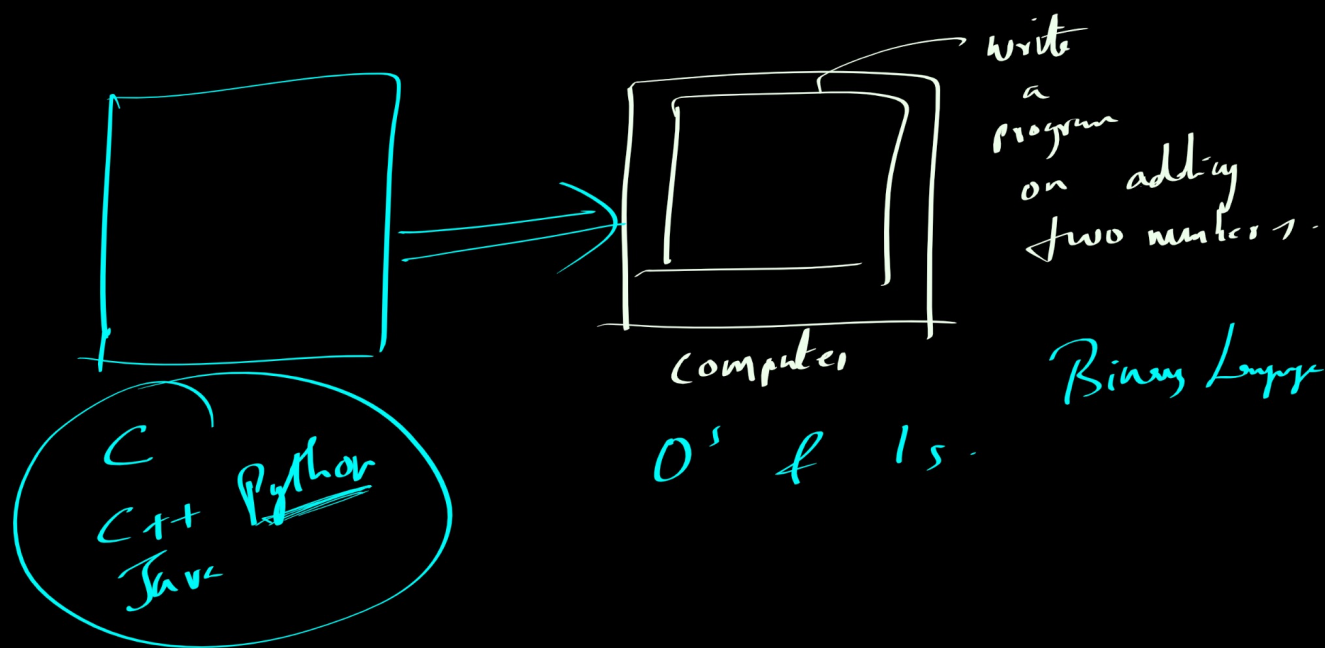
vs

Programming

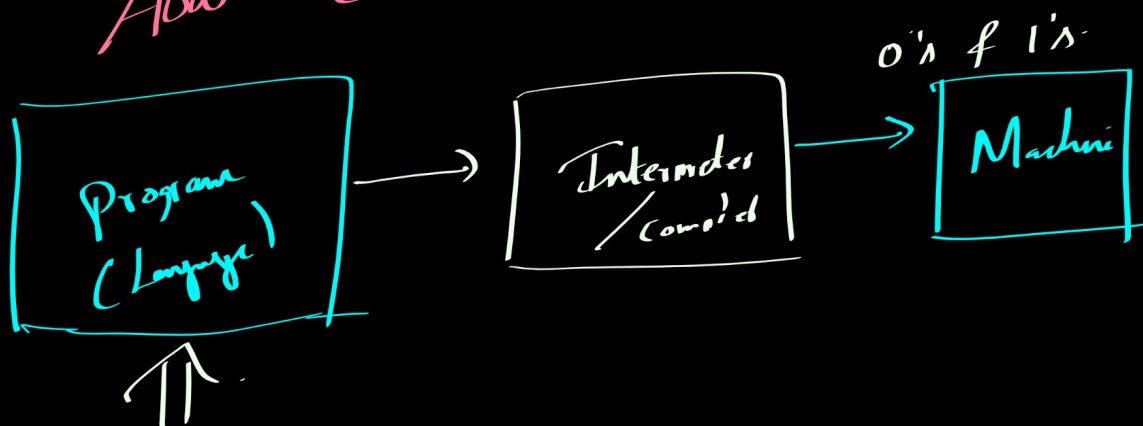
set of instructions



Programming Language (Python)
very easy

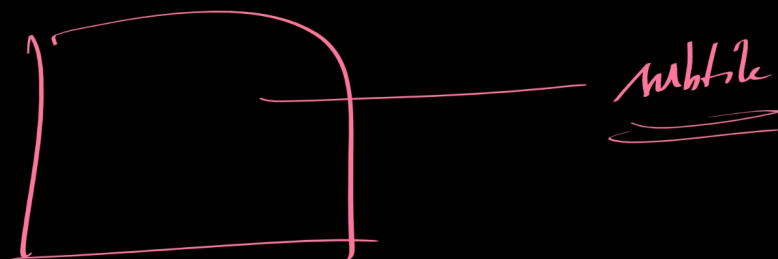
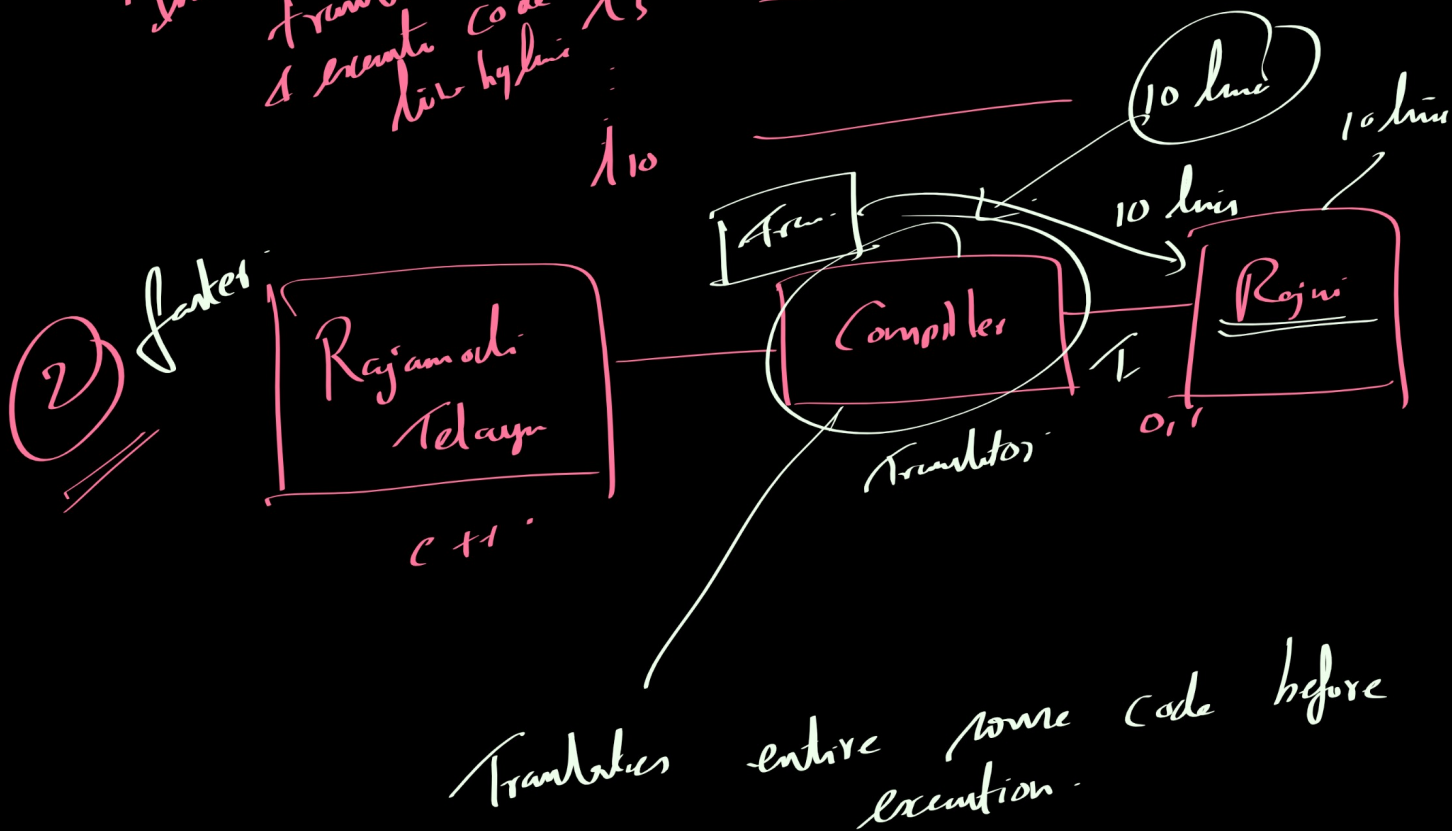
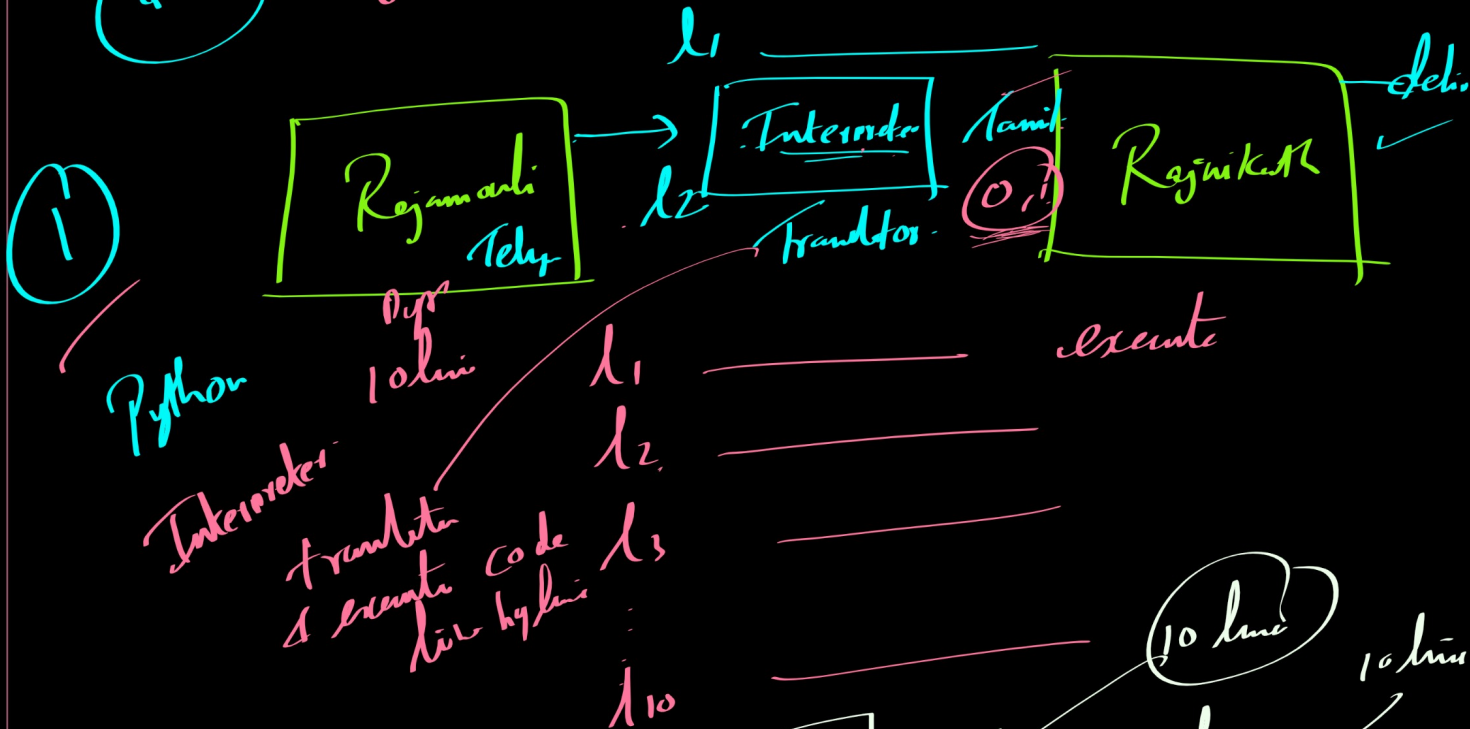


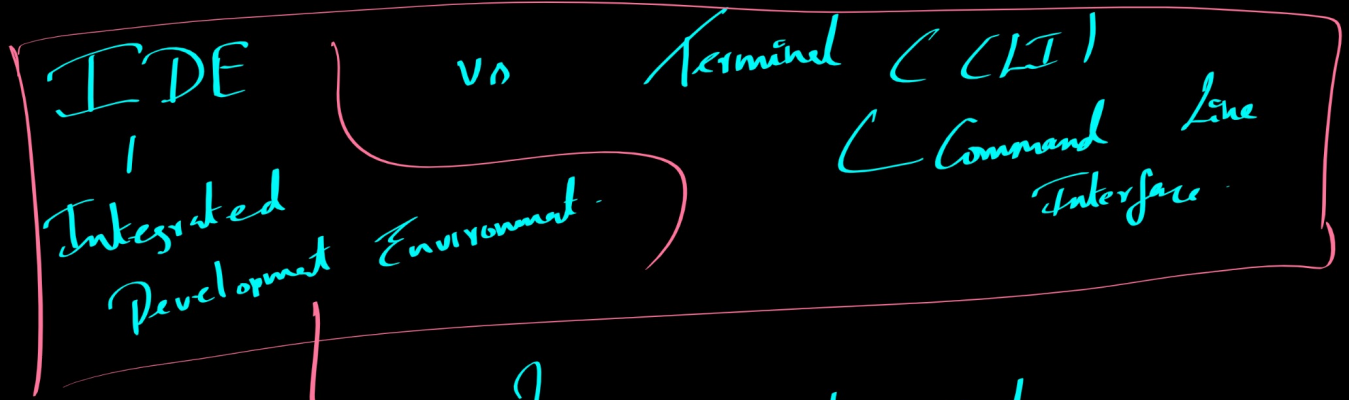
How code runs ?



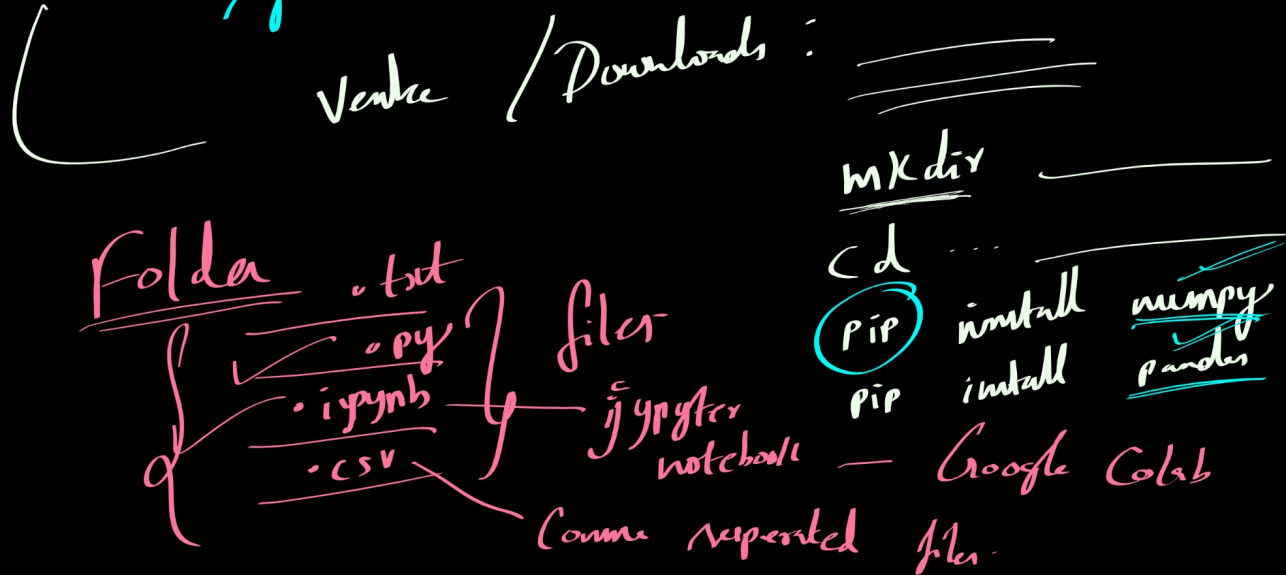
Rajini Kantli : Tamil
 Rajamouli : Telugu

10 lines

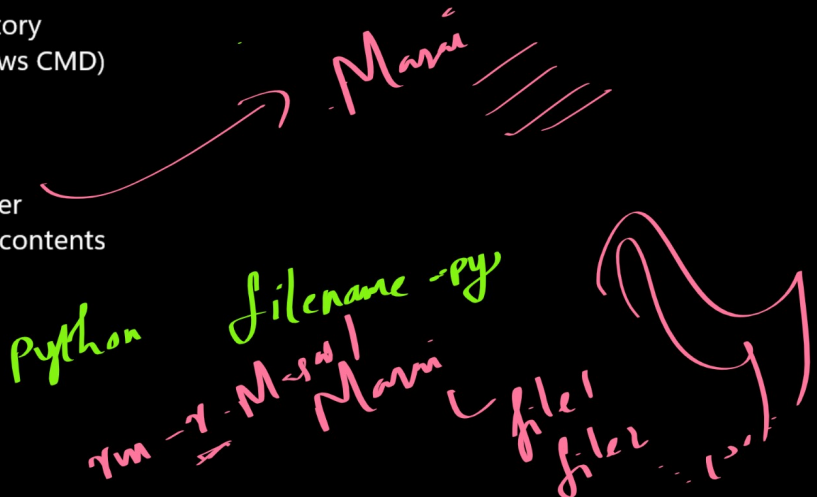




CLI: Direct Communication with the operating system using commands.



pwd # show current directory
ls # list files (dir in Windows CMD)
cd folder_name # go into folder
cd .. # go back
mkdir test # create folder
rmdir test # delete empty folder
rm -r test # delete folder with contents



Downloads
└─ Msui.html ✓
└─ Msui class 1.py ✓

mv filename path-to-push

render Msui
└─ Msui1
└─
└─
└─
└─

rm -r Msui2/
Msui2
files
files

rm filename

mv filename path
local h/c

test

Project 1

numpy == 1.21

1

Pip install numpy

Virtual environment

libraries
you
need

Project 2

numpy == 1.25

env 1

python

numpy

pandas

matplotlib

env 2

transformers

gensim

gensai

Virtual
environment

• env

secret
key

param

Google API

Google API

Application
Programming
Interface

